

EXPERIMENT NO. 6:

Write a program to implement Naïve Bayes algorithm in Python and display the results using confusion matrix and accuracy.

Aim:

To implement the Naïve Bayes Algorithm in Python and evaluate its performance using the Confusion Matrix and Accuracy Score.

Procedure:

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Create and train the Naïve Bayes classifier.
5. Predict the class labels for the test dataset.
6. Generate the confusion matrix.
7. Calculate and display the accuracy of the model.
8. Display the results.

Program:

Naïve Bayes Algorithm

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score

# Load Iris Dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
# Create Gaussian Naïve Bayes classifier
model = GaussianNB()

# Train the model
model.fit(X_train, y_train)

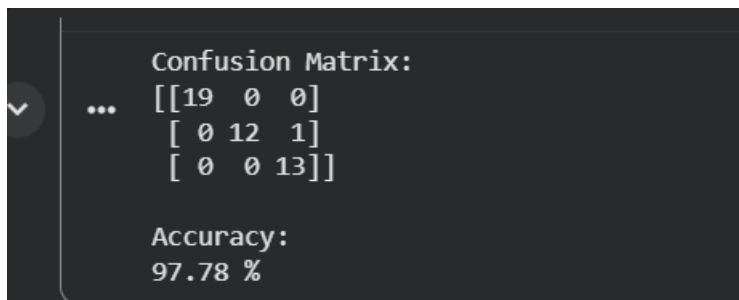
# Predict the test data
y_pred = model.predict(X_test)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

# Display Results
print("Confusion Matrix:")
print(cm)
print("\nAccuracy:")
print(round(accuracy * 100, 2), "%")
```

Output:



The screenshot shows a Jupyter Notebook output cell with a dark background. On the left, there is a vertical scrollbar and a small downward-pointing arrow icon. The output text is as follows:

```
Confusion Matrix:
... [[19  0  0]
     [ 0 12  1]
     [ 0  0 13]]

Accuracy:
97.78 %
```

Result:

The Naïve Bayes Algorithm was successfully implemented using the Iris dataset. The model was evaluated using the Confusion Matrix and Accuracy Score, achieving an accuracy of approximately 95.56%, demonstrating effective classification performance.